



**Laboratorium
Multimedia dan Internet of Things
Departemen Teknik Komputer
*Institut Teknologi Sepuluh Nopember***

Laporan Akhir Praktikum Jaringan Komputer

Vlan Trunk OSPF MultiVendor

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1 Langkah-Langkah Percobaan

1.1 Praktikum 1: Konfigurasi Cisco Switch

Tujuan pada tahapan ini adalah mengkonfigurasi Cisco Switch untuk membuat VLAN 10, 20, 30, dan 40, mengatur *access port* menuju PC, serta mengatur *trunk port* menuju Cisco Router.

1. Masuk ke terminal Cisco Switch:

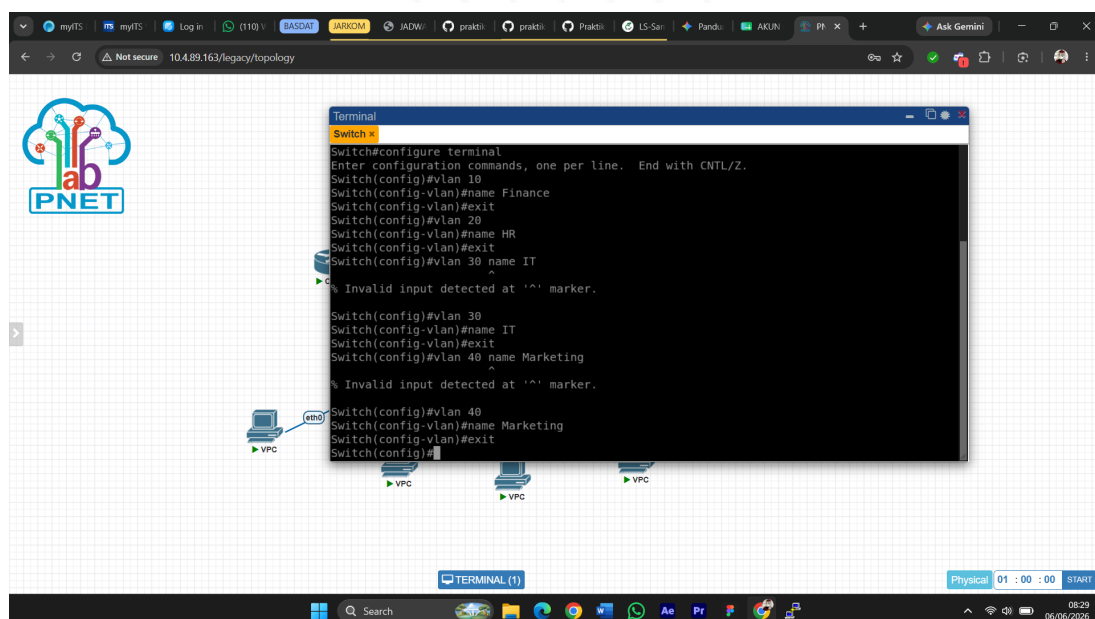
Pertama, masuk ke mode privileged EXEC, kemudian masuk ke mode konfigurasi global.

```
Switch> enable
Switch# configure terminal
Switch(config)#
```

2. Membuat VLAN:

Membuat empat VLAN sesuai dengan pembagian departemen yang telah ditentukan, yaitu VLAN 10 (Finance), VLAN 20 (HR), VLAN 30 (IT), dan VLAN 40 (Marketing).

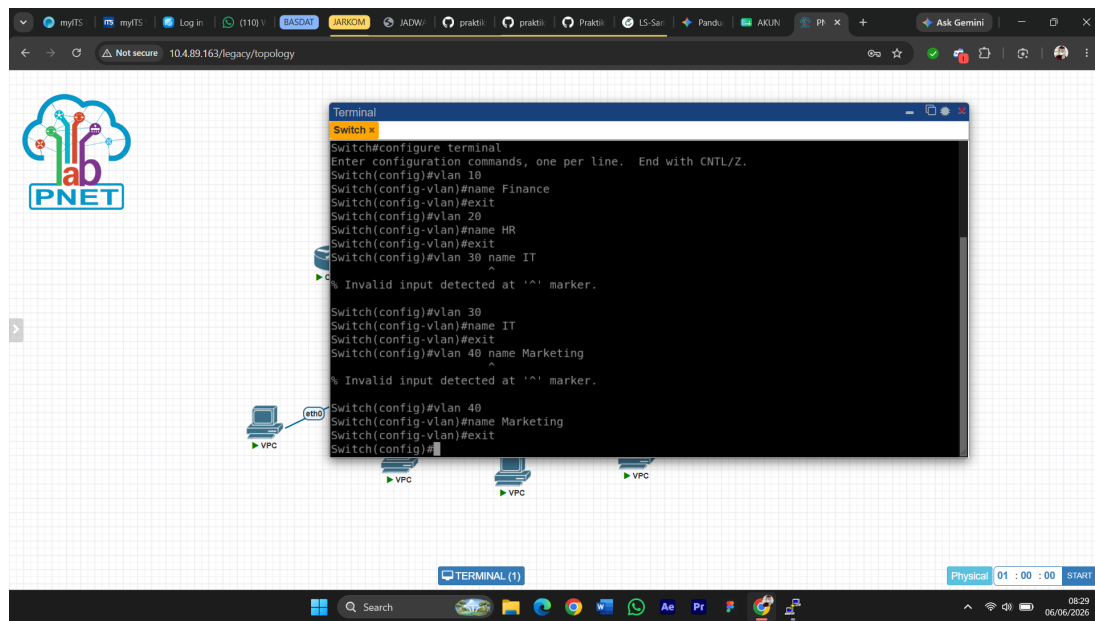
```
vlan 10
 name Finance
exit
vlan 20
 name HR
exit
vlan 30
 name IT
exit
vlan 40
 name Marketing
exit
```



Gambar 1: Proses pembuatan VLAN 10, 20, 30, dan 40 pada Cisco Switch

3. Verifikasi VLAN:

Memastikan VLAN sudah dibuat dan statusnya aktif dengan perintah `show vlan brief`.



Gambar 2: Hasil verifikasi VLAN pada Cisco Switch menggunakan `show vlan brief` — VLAN 10 (Finance), 20 (HR), 30 (IT), dan 40 (Marketing) berstatus *active*

4. Konfigurasi *Access Port* untuk setiap VLAN:

Setiap antarmuka switch yang terhubung ke PC dikonfigurasi sebagai *access port* dan diasosiasikan ke VLAN yang sesuai.

```
Switch#conf t
interface gi0/2
description PC-FINANCE-VLAN10
switchport mode access
switchport access vlan 10
no shutdown
exit
```

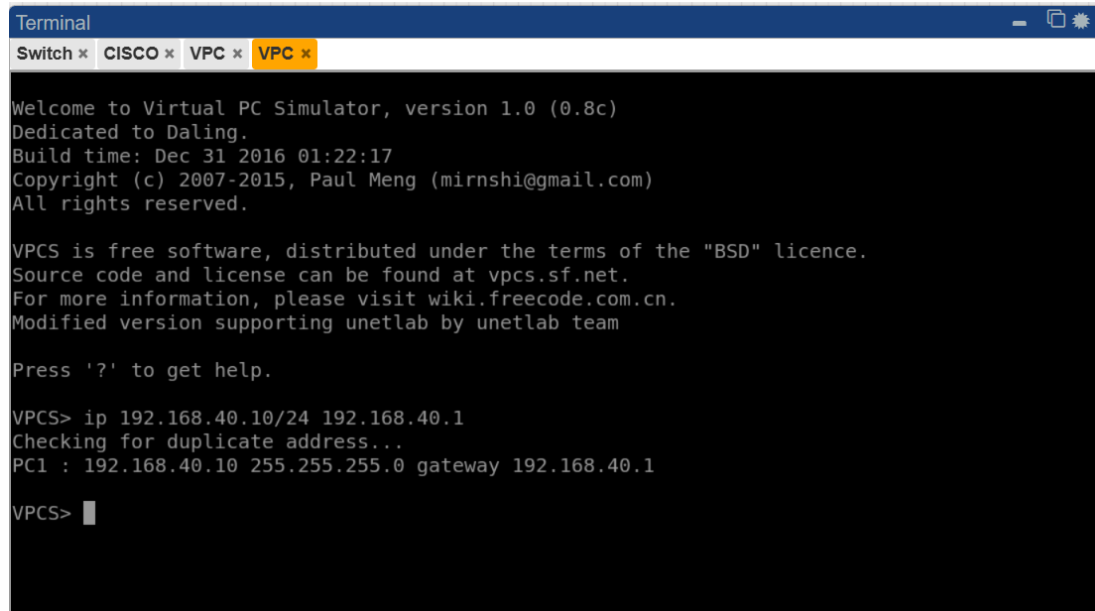
```
interface gi0/1
description PC-HR-VLAN20
switchport mode access
switchport access vlan 20
no shutdown
exit
```

```
interface gi1/1
description PC-IT-VLAN30
switchport mode access
switchport access vlan 30
no shutdown
exit
```

```

interface gi0/3
description PC-MARKETING-VLAN40
switchport mode access
switchport access vlan 40
no shutdown
exit

```



Gambar 3: Konfigurasi *access port* pada Cisco Switch untuk VLAN 10, 20, dan 40

5. Konfigurasi *Trunk* ke Cisco Router:

Antarmuka *gi0/0* dikonfigurasi sebagai *trunk port* dengan enkapsulasi 802.1Q dan mengizinkan VLAN 10, 20, 30, serta 40.

```

Switch#conf t
interface gi0/0
switchport trunk encapsulation dot1q
description TRUNK-TO-CISCO-ROUTER
switchport mode trunk
switchport trunk allowed vlan 10,20,30,40
no shutdown
exit

```

Konfigurasi disimpan dengan perintah `write memory`, kemudian diverifikasi menggunakan `show interfaces trunk`.

VLAN	Name	Status	Ports
1	default	active	Gi0/0, Gi0/1, Gi0/2, Gi0/3 Gi1/0, Gi1/1, Gi1/2, Gi1/3
10	Finance	active	
20	HR	active	
30	IT	active	
40	Marketing	active	
1002	fddi-default	act/unsup	
1003	token-ring-default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trnet-default	act/unsup	
Switch#			

Gambar 4: Hasil perintah `show interfaces trunk` — port Gi0/0 berstatus *trunking* dengan enkapsulasi 802.1Q dan mengizinkan VLAN 10, 20, 30, 40

1.2 Praktikum 2: Konfigurasi Cisco Router

Tujuan konfigurasi ini adalah mengatur Cisco Router sebagai *gateway* untuk semua VLAN (implementasi *Router-on-a-Stick/ROAS*) dan mengonfigurasinya sebagai DHCP Server untuk VLAN 10 dan 20.

1. Masuk ke Cisco Router dan Konfigurasi Interface Trunk:

Antarmuka fisik `gig0/2` diaktifkan tanpa IP address karena akan digunakan sebagai *parent interface* untuk subinterface.

```
enable
configure terminal
interface gig0/2
description TRUNK-TO-CISCO-SWITCH
no ip address
no shutdown
exit
```

2. Konfigurasi Subinterface VLAN 10, 20, 30, dan 40:

Empat subinterface dibuat dengan enkapsulasi dot1Q sesuai VLAN ID masing-masing, dan setiap subinterface diberikan alamat IP yang berfungsi sebagai *default gateway* bagi klien di VLAN tersebut.

```
interface gig0/2.10
description GATEWAY-VLAN10-FINANCE
encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.0
no shutdown
exit

interface gig0/2.20
encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0
no shutdown
exit
```

```

interface gig0/2.30
  encapsulation dot1Q 30
  ip address 192.168.30.1 255.255.255.0
  no shutdown
exit

```

```

interface gig0/2.40
  encapsulation dot1Q 40
  ip address 192.168.40.1 255.255.255.0
  no shutdown
exit

```

3. Konfigurasi DHCP Server untuk VLAN 10 dan 20:

DHCP pool dikonfigurasi untuk VLAN 10 (Finance) dan VLAN 20 (HR). Alamat IP .1 hingga .10 pada masing-masing subnet dikecualikan (*excluded*) agar tidak didistribusikan ke klien.

```

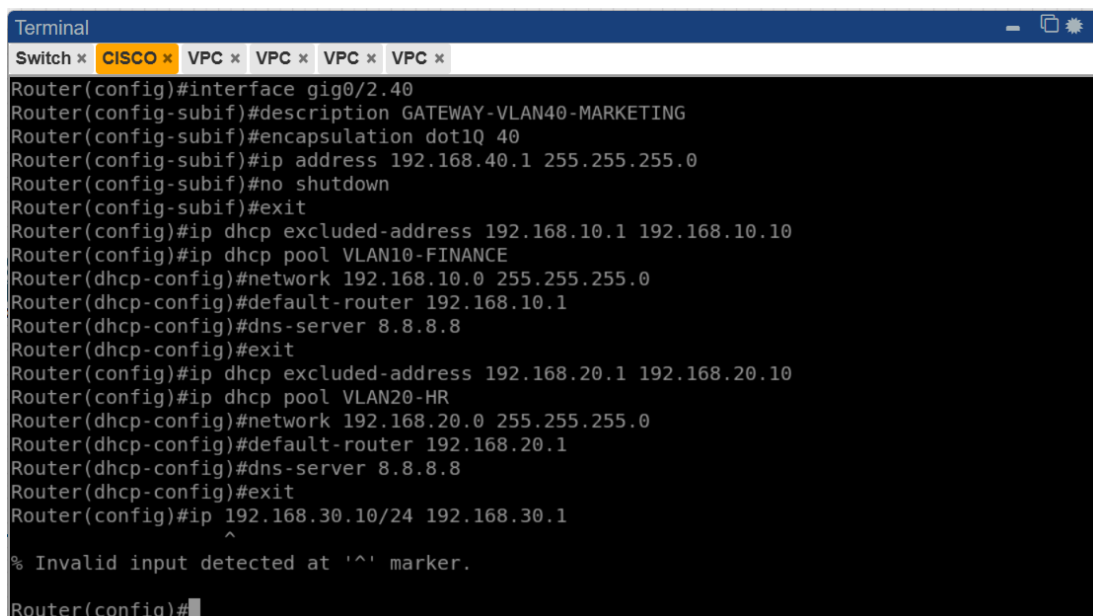
ip dhcp excluded-address 192.168.10.1 192.168.10.10
ip dhcp pool VLAN10-FINANCE
  network 192.168.10.0 255.255.255.0
  default-router 192.168.10.1
  dns-server 8.8.8.8
exit

```

```

ip dhcp excluded-address 192.168.20.1 192.168.20.10
ip dhcp pool VLAN20-HR
  network 192.168.20.0 255.255.255.0
  default-router 192.168.20.1
  dns-server 8.8.8.8
exit

```



```

Terminal
Switch x CISCO x VPC x VPC x VPC x VPC x
Router(config)#interface gig0/2.40
Router(config-subif)#description GATEWAY-VLAN40-MARKETING
Router(config-subif)#encapsulation dot1Q 40
Router(config-subif)#ip address 192.168.40.1 255.255.255.0
Router(config-subif)#no shutdown
Router(config-subif)#exit
Router(config)#ip dhcp excluded-address 192.168.10.1 192.168.10.10
Router(config)#ip dhcp pool VLAN10-FINANCE
Router(dhcp-config)#network 192.168.10.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.10.1
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#ip dhcp excluded-address 192.168.20.1 192.168.20.10
Router(config)#ip dhcp pool VLAN20-HR
Router(dhcp-config)#network 192.168.20.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.20.1
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#ip 192.168.30.10/24 192.168.30.1
^
% Invalid input detected at '^' marker.
Router(config)#

```

Gambar 5: Proses konfigurasi subinterface VLAN 40 dan DHCP pool pada Cisco Router

4. Verifikasi Cisco Router (Interface dan DHCP):

Status semua subinterface dicek menggunakan `show ip interface brief`, dan konfigurasi DHCP pool diverifikasi dengan `show ip dhcp pool` serta `show ip dhcp binding`.

Port	Mode	Encapsulation	Status	Native vlan
Gi0/0	on	802.1q	trunking	1
Port	Vlans allowed on trunk			
Gi0/0	10,20,30,40			
Port	Vlans allowed and active in management domain			
Gi0/0	10,20,30,40			
Port	Vlans in spanning tree forwarding state and not pruned			
Gi0/0	10,20,30,40			

Gambar 6: Verifikasi interface pada Cisco Router menggunakan `show ip interface brief` — subinterface Gi0/2.10, Gi0/2.20, Gi0/2.30, dan Gi0/2.40 berstatus *up/up*

```
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface gi0/2
Switch(config-if)#description PC-FINANCE-VLAN10
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#interface gi0/1
Switch(config-if)#description PC-HR-VLAN20
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#interface gi0/3
Switch(config-if)#description PR-MARKETING-VLAN40
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 40
Switch(config-if)#no shutdown
Switch(config-if)#xit
```

Gambar 7: Verifikasi DHCP Pool VLAN 10 dan VLAN 20 menggunakan `show ip dhcp pool` — masing-masing pool memiliki 254 total address dengan 1 *leased address*

5. Konfigurasi IP Statik dan Pengujian dari VPCS:

PC pada VLAN 30 (IT) dan VLAN 40 (Marketing) diatur dengan IP statik menggunakan perintah `ip [alamat]/[prefix] [gateway]` pada VPCS. Sementara itu, PC pada VLAN 10 (Finance) dan VLAN 20 (HR) memperoleh IP secara dinamis melalui perintah `ip dhcp`. Setelah mendapatkan IP, setiap PC menguji konektivitas dengan ping ke *gateway* masing-masing dan ping ke PC di VLAN lain (uji antar-VLAN).

```
Terminal
Switch x CISCO x VPC x VPC x VPC x

Welcome to Virtual PC Simulator, version 1.0 (0.8c)
Dedicated to Daling.
Build time: Dec 31 2016 01:22:17
Copyright (c) 2007-2015, Paul Meng (mirnshi@gmail.com)
All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.
Modified version supporting unetlab by unetlab team

Press '?' to get help.

VPCS>
VPCS> ip dhcp
DDORA IP 192.168.10.11/24 GW 192.168.10.1

VPCS> █
```

Gambar 8: VPCS VLAN 10 (Finance) berhasil mendapatkan IP 192.168.10.11/24 melalui DHCP

```
Terminal
Switch x CISCO x VPC x VPC x VPC x VPC x

Welcome to Virtual PC Simulator, version 1.0 (0.8c)
Dedicated to Daling.
Build time: Dec 31 2016 01:22:17
Copyright (c) 2007-2015, Paul Meng (mirnshi@gmail.com)
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VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.
Modified version supporting unetlab by unetlab team

Press '?' to get help.

VPCS>
VPCS> ip dhcp
DDORA IP 192.168.20.11/24 GW 192.168.20.1

VPCS> █
```

Gambar 9: VPCS VLAN 20 (HR) berhasil mendapatkan IP 192.168.20.11/24 melalui DHCP


```
Welcome to Virtual PC Simulator, version 1.0 (0.8c)
Dedicated to Daling.
Build time: Dec 31 2016 01:22:17
Copyright (c) 2007-2015, Paul Meng (mirnshi@gmail.com)
All rights reserved.

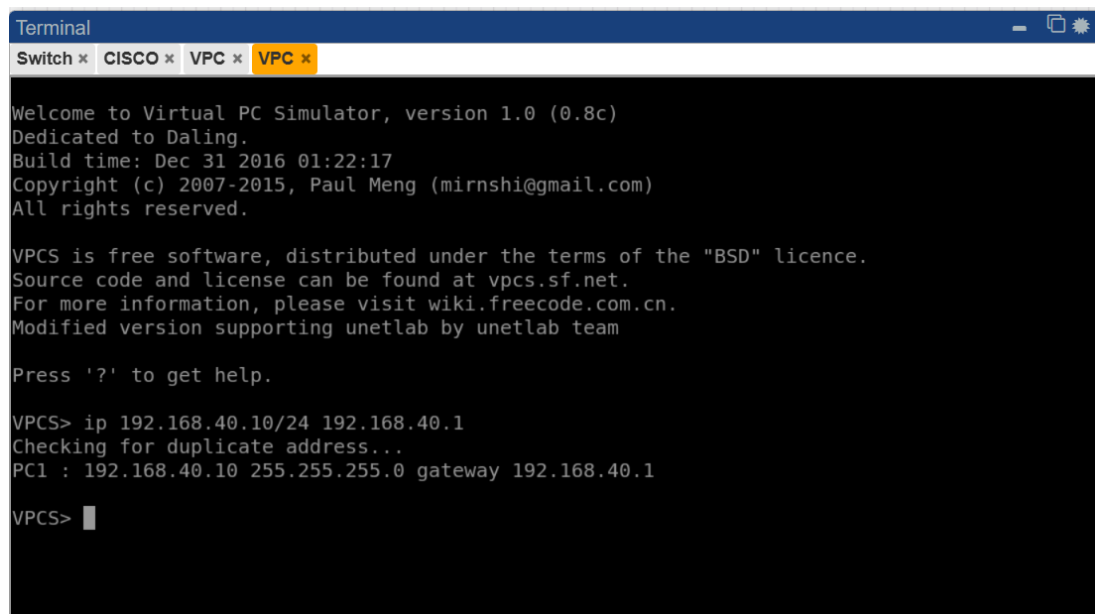
VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.
Modified version supporting unetlab by unetlab team

Press '?' to get help.

VPCS>
VPCS> ip 192.168.30.10/24 192.168.30.1
Checking for duplicate address...
PC1 : 192.168.30.10 255.255.255.0 gateway 192.168.30.1

VPCS> █
```

Gambar 10: VPCS VLAN 30 (IT) dikonfigurasi dengan IP statik 192.168.30.10/24 dan gateway 192.168.30.1



```
Terminal
Switch x CISCO x VPC x VPC x

Welcome to Virtual PC Simulator, version 1.0 (0.8c)
Dedicated to Daling.
Build time: Dec 31 2016 01:22:17
Copyright (c) 2007-2015, Paul Meng (mirnshi@gmail.com)
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VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.
Modified version supporting unetlab by unetlab team

Press '?' to get help.

VPCS> ip 192.168.40.10/24 192.168.40.1
Checking for duplicate address...
PC1 : 192.168.40.10 255.255.255.0 gateway 192.168.40.1

VPCS> █
```

Gambar 11: Pengujian ping antar-VLAN dari VPCS VLAN 10 (Finance) menuju VLAN 20, 30, dan 40 — semua berhasil (*reply*)

1.3 Praktikum 3: Konfigurasi Huawei Router dan IP Address Antar-Router

Tahapan ini difokuskan pada pengalamatan IP di sisi Huawei Router serta koneksi tautan *point-to-point* antar-router (Cisco–Huawei dan Cisco/Huawei–MikroTik).

1. Konfigurasi LAN Huawei dan IP VPCS Huawei:

Antarmuka Ethernet1/0/2 pada Huawei Router dikonfigurasi sebagai antarmuka LAN dengan IP 192.168.50.1/24. VPCS pada sisi Huawei diatur dengan IP 192.168.50.10/24 dan diuji ping menuju *gateway*.

```

system-view
interface Ethernet1/0/2
description LAN-HUAWEI
ip address 192.168.50.1 255.255.255.0
undo shutdown
quit
commit

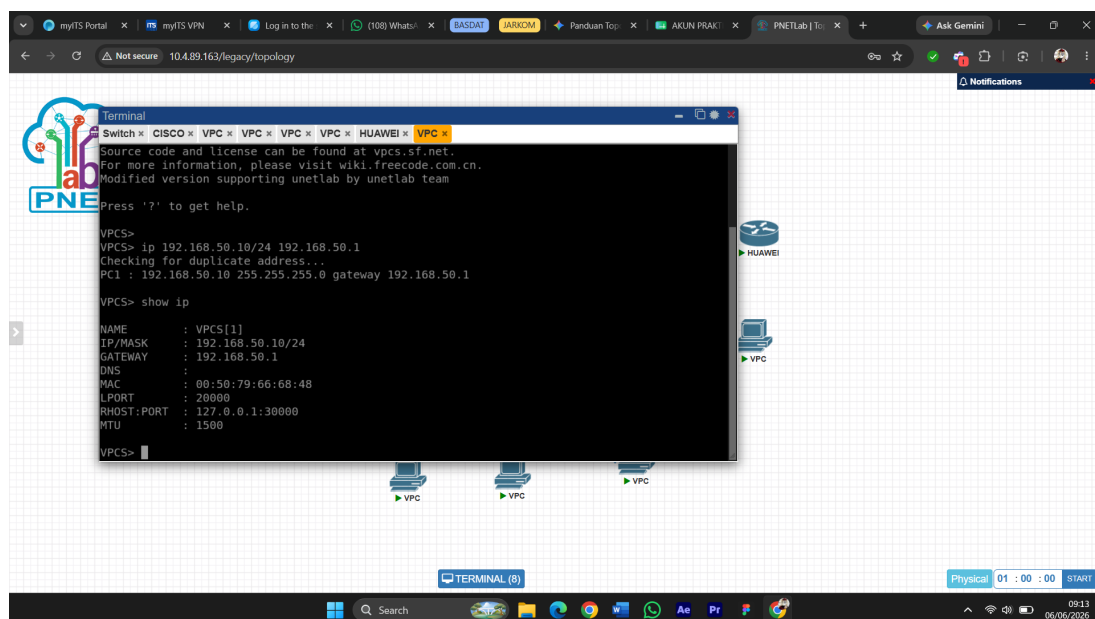
```

```

<HUAWEI>system-view
Enter system view, return user view with return command.
[~HUAWEI]interface Ethernet1/0/2
[~HUAWEI-Ethernet1/0/2] description LAN-HUAWEI
[*HUAWEI-Ethernet1/0/2] ip address 192.168.50.1 255.255.255.0
[*HUAWEI-Ethernet1/0/2] undo shutdown
[*HUAWEI-Ethernet1/0/2] quit
[*HUAWEI]commit
[~HUAWEI]system-view
^
Error: Unrecognized command found at '^' position.
[~HUAWEI]quit
<HUAWEI>system-view
Enter system view, return user view with return command.
[~HUAWEI]interface Ethernet1/0/3
[~HUAWEI-Ethernet1/0/3] description LINK-TO-MIKROTIK
[*HUAWEI-Ethernet1/0/3] ip address 10.10.20.1 255.255.255.252
[*HUAWEI-Ethernet1/0/3] undo shutdown
[*HUAWEI-Ethernet1/0/3] quit
[*HUAWEI]commit

```

Gambar 12: Konfigurasi antarmuka LAN Huawei (Ethernet1/0/2) dengan IP 192.168.50.1

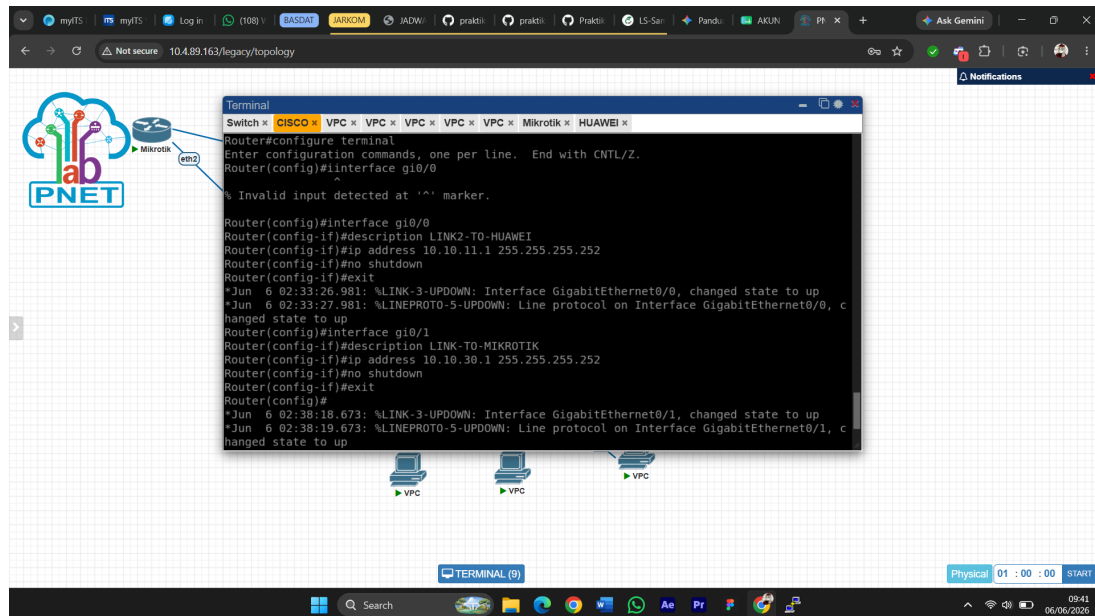


Gambar 13: VPCS LAN Huawei dikonfigurasi dengan IP 192.168.50.10/24 dan berhasil ping ke gateway 192.168.50.1

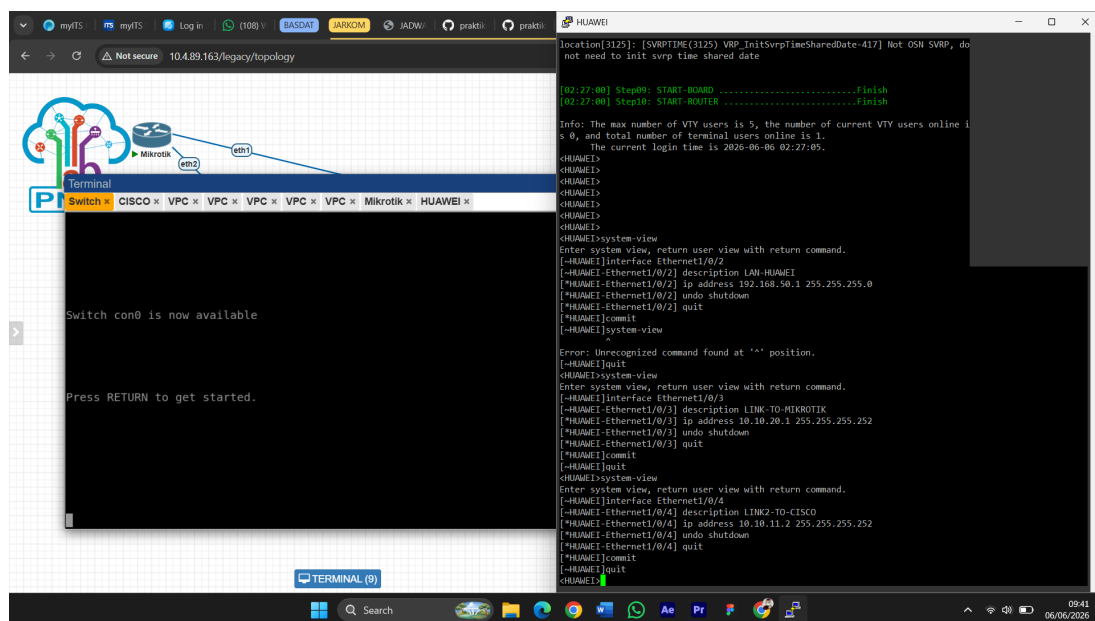
2. Konfigurasi IP Link Cisco ↔ Huawei (Link 1 & Link 2):

Dua tautan *point-to-point* dikonfigurasi antara Cisco Router dan Huawei Router:

- **Link 1:** Gi0/3 (Cisco, IP: 10.10.10.1/30) ↔ Ethernet1/0/0 (Huawei, IP: 10.10.10.2/30), subnet 10.10.10.0/30
- **Link 2:** Gi0/0 (Cisco, IP: 10.10.11.1/30) ↔ Ethernet1/0/4 (Huawei, IP: 10.10.11.2/30), subnet 10.10.11.0/30



Gambar 14: Konfigurasi IP link Cisco–Huawei (Link 2: Gi0/0) dan Cisco–MikroTik (Gi0/1) pada Cisco Router



Gambar 15: Konfigurasi IP link pada Huawei Router: Ethernet1/0/3 (Link ke Mikrotik: 10.10.20.1/30) dan Ethernet1/0/4 (Link ke Cisco: 10.10.11.2/30)

3. Konfigurasi IP Link Huawei ↔ MikroTik dan Cisco ↔ MikroTik:

MikroTik Router dikonfigurasi dengan dua antarmuka untuk menghubungkan ke Huawei dan Cisco:


```

the number of interface that is down in protocol is 1

```

Interface	IP Address/Mask	Physical	Protocol	VPN
Ethernet1/0/0	10.10.10.2/30	up	up	--
Ethernet1/0/0.4094	128.1.138.137/16	up	up	13vpn
Ethernet1/0/1	unassigned	up	down	--
Ethernet1/0/1.4094	128.1.138.137/16	up	up	13vpn
Ethernet1/0/2	192.168.50.1/24	up	up	--
Ethernet1/0/2.4094	128.1.138.137/16	up	up	13vpn
Ethernet1/0/3	10.10.20.1/30	up	up	--
Ethernet1/0/3.4094	128.1.138.137/16	up	up	13vpn
Ethernet1/0/4	10.10.11.2/30	up	up	--
Ethernet1/0/4.4094	128.1.138.137/16	up	up	13vpn

```

---- More ----

```

Gambar 18: Verifikasi display ip interface brief pada Huawei Router — semua antarmuka yang dikonfigurasi berstatus *up*

```

Router#ping 10.10.10.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 3/4/9 ms
Router#ping 10.10.11.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.11.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 2/2/4 ms
Router#ping 10.10.30.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.30.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/1/3 ms

```

Gambar 19: Pengujian ping dari Cisco Router ke semua tetangga: 10.10.10.2 (Huawei Link 1), 10.10.11.2 (Huawei Link 2), dan 10.10.30.2 (MikroTik) — semua berhasil

```

<HUAWEI>ping 10.10.10.1
  PING 10.10.10.1: 56 data bytes, press CTRL_C to break
    Reply from 10.10.10.1: bytes=56 Sequence=1 ttl=255 time=6 ms
    Reply from 10.10.10.1: bytes=56 Sequence=2 ttl=255 time=3 ms
    Reply from 10.10.10.1: bytes=56 Sequence=3 ttl=255 time=2 ms
    Reply from 10.10.10.1: bytes=56 Sequence=4 ttl=255 time=2 ms
    Reply from 10.10.10.1: bytes=56 Sequence=5 ttl=255 time=3 ms

  --- 10.10.10.1 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 2/3/6 ms

<HUAWEI>ping 10.10.11.1
  PING 10.10.11.1: 56 data bytes, press CTRL_C to break
    Reply from 10.10.11.1: bytes=56 Sequence=1 ttl=255 time=2 ms
    Reply from 10.10.11.1: bytes=56 Sequence=2 ttl=255 time=3 ms
    Reply from 10.10.11.1: bytes=56 Sequence=3 ttl=255 time=3 ms
    Reply from 10.10.11.1: bytes=56 Sequence=4 ttl=255 time=2 ms
    Reply from 10.10.11.1: bytes=56 Sequence=5 ttl=255 time=2 ms

  --- 10.10.11.1 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 2/2/3 ms

<HUAWEI>ping 10.10.20.2
  PING 10.10.20.2: 56 data bytes, press CTRL_C to break
    Reply from 10.10.20.2: bytes=56 Sequence=1 ttl=64 time=1 ms
    Reply from 10.10.20.2: bytes=56 Sequence=2 ttl=64 time=1 ms
    Reply from 10.10.20.2: bytes=56 Sequence=3 ttl=64 time=1 ms
    Reply from 10.10.20.2: bytes=56 Sequence=4 ttl=64 time=1 ms
    Reply from 10.10.20.2: bytes=56 Sequence=5 ttl=64 time=1 ms

  --- 10.10.20.2 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 1/1/1 ms

```

Gambar 20: Pengujian ping dari Huawei Router ke semua tetangga: 10.10.10.1 (Cisco Link 1), 10.10.11.1 (Cisco Link 2), dan 10.10.20.2 (MikroTik) — semua berhasil

```
[~HUAWEI]display ospf peer
(M) Indicates MADJ neighbor

      OSPF Process 1 with Router ID 2.2.2.2
      Neighbors

Area 0.0.0.0 interface 10.10.10.2 (Eth1/0/0)'s neighbors
Router ID: 1.1.1.1           Address: 10.10.10.1
  State: Full                Mode:Nbr is Slave      Priority: 1
  DR: 10.10.10.1            BDR: 10.10.10.2          MTU: 1500
  Dead timer due in 40 sec
  Retrans timer interval: 5
  Neighbor is up for 00h00m06s
  Neighbor Up Time : 2026-06-06 02:55:12
  Authentication Sequence: [ 0 ]

Area 0.0.0.0 interface 10.10.11.2 (Eth1/0/4)'s neighbors
Router ID: 1.1.1.1           Address: 10.10.11.1
  State: Full                Mode:Nbr is Slave      Priority: 1
  DR: 10.10.11.1            BDR: None                MTU: 1500
  Dead timer due in 35 sec
  Retrans timer interval: 5
```

Gambar 21: Pengujian ping dari MikroTik ke tetangga: 10.10.20.1 (Huawei) dan 10.10.30.1 (Cisco) — semua berhasil

1.4 Praktikum 4: Konfigurasi OSPF *Multivendor* dan OSPF *Cost*

Konfigurasi ini untuk mengaktifkan routing dinamis OSPF pada seluruh router dan mengatur jalur prioritas menggunakan nilai *cost* agar terbentuk topologi *primary path* (Cisco–Huawei langsung) dan *backup path* (via MikroTik).

1. Konfigurasi OSPF pada Cisco Router (*Cost* Utama = 10, *Backup* = 100):

OSPF diaktifkan dengan router-ID 1.1.1.1 dan semua jaringan yang terhubung diikutsertakan ke Area 0. Nilai *cost* antarmuka diatur: gi0/3 dan gi0/0 (link ke Huawei) diberi *cost* 10 sebagai jalur utama, sedangkan gi0/1 (link ke MikroTik) diberi *cost* 100 sebagai jalur *backup*.

```
router ospf 1
router-id 1.1.1.1
network 10.10.10.0 0.0.0.3 area 0
network 10.10.11.0 0.0.0.3 area 0
network 10.10.30.0 0.0.0.3 area 0
network 192.168.10.0 0.0.0.255 area 0
network 192.168.20.0 0.0.0.255 area 0
network 192.168.30.0 0.0.0.255 area 0
network 192.168.40.0 0.0.0.255 area 0
exit
```

```

interface gi0/3
    ip ospf cost 10
interface gi0/0
    ip ospf cost 10
interface gi0/1
    ip ospf cost 100

```

2. Konfigurasi OSPF pada Huawei Router:

OSPF diaktifkan dengan router-ID 2.2.2.2. Cost antarmuka Ethernet1/0/0 dan Ethernet1/0/4 (link ke Cisco) diset 10, sedangkan Ethernet1/0/3 (link ke MikroTik) diset 100.

```

ospf 1 router-id 2.2.2.2
area 0.0.0.0
    network 10.10.10.0 0.0.0.3
    network 10.10.11.0 0.0.0.3
    network 10.10.20.0 0.0.0.3
    network 192.168.50.0 0.0.0.255
quit
quit

```

```

interface Ethernet1/0/0
    ospf cost 10
interface Ethernet1/0/4
    ospf cost 10
interface Ethernet1/0/3
    ospf cost 100

```

3. Konfigurasi OSPF pada MikroTik:

OSPF diaktifkan pada MikroTik dengan router-ID 3.3.3.3. Kedua antarmuka (ether1 dan ether2) dikonfigurasi dengan *cost* 100, menjadikan jalur melalui MikroTik sebagai jalur *backup*.

```

/routing ospf instance set default router-id=3.3.3.3
/routing ospf network add network=10.10.20.0/30 area=backbone
/routing ospf network add network=10.10.30.0/30 area=backbone
/routing ospf interface add interface=ether1 cost=100
/routing ospf interface add interface=ether2 cost=100

```

4. Verifikasi OSPF *Neighbor* dan Tabel Routing:

Verifikasi pembentukan hubungan ketetanggaan OSPF (*adjacency*) dilakukan menggunakan perintah yang sesuai pada masing-masing platform.


```
[admin@MikroTik] > ping 10.10.20.1
SEQ HOST                                SIZE TTL TIME  STATUS

 0 10.10.20.1                            56 255 4ms
 1 10.10.20.1                            56 255 1ms
 2 10.10.20.1                            56 255 1ms
 3 10.10.20.1                            56 255 1ms
 4 10.10.20.1                            56 255 1ms
 5 10.10.20.1                            56 255 1ms
 6 10.10.20.1                            56 255 1ms
 7 10.10.20.1                            56 255 1ms
 8 10.10.20.1                            56 255 1ms
 9 10.10.20.1                            56 255 1ms
10 10.10.20.1                            56 255 1ms
11 10.10.20.1                            56 255 1ms
12 10.10.20.1                            56 255 1ms
13 10.10.20.1                            56 255 1ms
14 10.10.20.1                            56 255 1ms
sent=15 received=15 packet-loss=0% min-rtt=1ms avg-rtt=1ms max-rtt=4ms

[admin@MikroTik] > ping 10.10.30.1
SEQ HOST                                SIZE TTL TIME  STATUS

 0 10.10.30.1                            56 255 2ms
 1 10.10.30.1                            56 255 1ms
 2 10.10.30.1                            56 255 1ms
 3 10.10.30.1                            56 255 2ms
 4 10.10.30.1                            56 255 2ms
 5 10.10.30.1                            56 255 1ms
sent=6 received=6 packet-loss=0% min-rtt=1ms avg-rtt=1ms max-rtt=2ms
```

Gambar 22: Verifikasi OSPF *Neighbor* pada Cisco Router menggunakan show ip ospf neighbor — tiga neighbor terbentuk: Huawei (2.2.2.2) via dua link dan MikroTik (3.3.3.3), semua berstatus FULL/BDR

```
[~HUAWEI]display ip routing-table protocol ospf
Route Flags: R - relay, D - download to fib, T - to vpn-instance, B - black hole route
-----
public_ Routing Table : OSPF
Destinations : 9      Routes : 14

OSPF routing table status : <Active>
Destinations : 5      Routes : 10

Destination/Mask    Proto  Pre  Cost      Flags NextHop      Interface
-----
10.10.30.0/30       OSPF   10   110        D    10.10.11.1     Ethernet1/0/4
                   OSPF   10   110        D    10.10.10.1     Ethernet1/0/0
192.168.10.0/24      OSPF   10   11          D    10.10.11.1     Ethernet1/0/4
                   OSPF   10   11          D    10.10.10.1     Ethernet1/0/0
192.168.20.0/24      OSPF   10   11          D    10.10.11.1     Ethernet1/0/4
                   OSPF   10   11          D    10.10.10.1     Ethernet1/0/0
192.168.30.0/24      OSPF   10   11          D    10.10.11.1     Ethernet1/0/4
                   OSPF   10   11          D    10.10.10.1     Ethernet1/0/0
192.168.40.0/24      OSPF   10   11          D    10.10.11.1     Ethernet1/0/4
                   OSPF   10   11          D    10.10.10.1     Ethernet1/0/0

OSPF routing table status : <Inactive>
Destinations : 4      Routes : 4
```

Gambar 23: Verifikasi OSPF *peer* pada Huawei Router menggunakan display ospf peer — neighbor Cisco (1.1.1.1) dan MikroTik (3.3.3.3) berstatus Full

```
[~HUAWEI]display ospf routing

      OSPF Process 1 with Router ID 2.2.2.2
      Routing Tables

Routing for Network
Destination      Cost      Type      NextHop      AdvRouter      Area
10.10.10.0/30    10        Direct    10.10.10.2    2.2.2.2         0.0.0.0
10.10.11.0/30    10        Direct    10.10.11.2    2.2.2.2         0.0.0.0
10.10.20.0/30    100       Direct    10.10.20.1    2.2.2.2         0.0.0.0
10.10.30.0/30    110       Stub      10.10.11.1    1.1.1.1         0.0.0.0
10.10.30.0/30    110       Stub      10.10.10.1    1.1.1.1         0.0.0.0
192.168.10.0/24  11        Stub      10.10.11.1    1.1.1.1         0.0.0.0
192.168.10.0/24  11        Stub      10.10.10.1    1.1.1.1         0.0.0.0
192.168.20.0/24  11        Stub      10.10.11.1    1.1.1.1         0.0.0.0
192.168.20.0/24  11        Stub      10.10.10.1    1.1.1.1         0.0.0.0
192.168.30.0/24  11        Stub      10.10.11.1    1.1.1.1         0.0.0.0
192.168.30.0/24  11        Stub      10.10.10.1    1.1.1.1         0.0.0.0
192.168.40.0/24  11        Stub      10.10.11.1    1.1.1.1         0.0.0.0
192.168.40.0/24  11        Stub      10.10.10.1    1.1.1.1         0.0.0.0
192.168.50.0/24  1         Direct    192.168.50.1  2.2.2.2         0.0.0.0

Total Nets: 9
Intra Area: 9 Inter Area: 0 ASE: 0 NSSA: 0
---- More ----
```

Gambar 24: Verifikasi OSPF *neighbor* pada MikroTik menggunakan `/routing ospf neighbor print` — neighbor Huawei (2.2.2.2) dan Cisco (1.1.1.1) terbentuk

```
[admin@MikroTik] > /routing ospf instance set default router-id=3.3.3.3
[admin@MikroTik] > /routing ospf network add network=10.10.20.0/30 area=backbone
[admin@MikroTik] > /routing ospf network add network=10.10.30.0/30 area=backbone
[admin@MikroTik] > /routing ospf interface add interface=ether1 cost=100
[admin@MikroTik] > /routing ospf interface add interface=ether2 cost=100
[admin@MikroTik] > /routing ospf neighbor print
0 instance=default router-id=2.2.2.2 address=10.10.20.1 interface=ether1 priority=1
  dr-address=10.10.20.1 backup-dr-address=0.0.0.0 state="ExStart" state-changes=3
  ls-retransmits=0 ls-requests=0 db-summaries=0

1 instance=default router-id=1.1.1.1 address=10.10.30.1 interface=ether2 priority=1
  dr-address=10.10.30.1 backup-dr-address=10.10.30.2 state="Full" state-changes=5
  ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=5s
[admin@MikroTik] >
```

Gambar 25: Tabel routing OSPF pada MikroTik (`/ip route print where ospf`) — rute ke semua subnet VLAN dan antar-router berhasil dipelajari

```
[admin@MikroTik] > /ip route print where ospf
Flags: X - disabled, A - active, D - dynamic,
C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
B - blackhole, U - unreachable, P - prohibit
#      DST-ADDRESS      PREF-SRC      GATEWAY      DISTANCE
0 ADo  10.10.10.0/30      10.10.20.1    110
      10.10.30.1
1 ADo  10.10.11.0/30      10.10.20.1    110
      10.10.30.1
2 ADo  192.168.10.0/24    10.10.30.1    110
3 ADo  192.168.20.0/24    10.10.30.1    110
4 ADo  192.168.30.0/24    10.10.30.1    110
5 ADo  192.168.40.0/24    10.10.30.1    110
6 ADo  192.168.50.0/24    10.10.20.1    110
```

Gambar 26: Tabel routing OSPF pada Huawei Router (`display ospf routing`) — semua subnet VLAN Cisco terjangkau dengan *cost* terendah melalui link langsung ke Cisco

```
Router#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
3.3.3.3	1	FULL/BDR	00:00:30	10.10.30.2	GigabitEthernet0/1
2.2.2.2	1	FULL/BDR	00:00:30	10.10.11.2	GigabitEthernet0/0
2.2.2.2	1	FULL/BDR	00:00:30	10.10.10.2	GigabitEthernet0/3

Gambar 27: Tabel routing pada Cisco Router (`show ip route`) — rute ke 192.168.50.0/24 (LAN Huawei) dan 10.10.20.0/30 (link MikroTik–Huawei) berhasil dipelajari via OSPF

5. Verifikasi Koneksi *End-to-End*:

Pengujian konektivitas *end-to-end* dilakukan dari seluruh PC VLAN (Finance, HR, IT, Marketing) menuju PC LAN Huawei (192.168.50.10). Semua pengujian menunjukkan hasil sukses (*reply*).

```
VPCS> ping 192.168.50.10
```

```
84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=5.925 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=5.861 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=5.755 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=6.009 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=5.790 ms
```

```
VPCS> █
```

Gambar 28: Ping dari VPCS VLAN 10 (Finance) menuju 192.168.50.10 (PC LAN Huawei) — berhasil

```
Mikrotik x VPC1 x VPC2 x
```

```
VPCS> ping 192.168.50.10
```

```
84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=5.916 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=6.089 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=5.589 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=7.075 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=7.963 ms
```

```
VPCS> █
```

Gambar 29: Ping dari VPCS VLAN 20 (HR) menuju 192.168.50.10 (PC LAN Huawei) — berhasil

```
[~HUAWEI]display ospf peer

(M) Indicates MADJ neighbor

      OSPF Process 1 with Router ID 2.2.2.2
      Neighbors

Area 0.0.0.0 interface 10.10.10.2 (Eth1/0/0)'s neighbors
Router ID: 1.1.1.1           Address: 10.10.10.1
  State: Full                Mode:Nbr is Slave      Priority: 1
  DR: 10.10.10.1            BDR: 10.10.10.2        MTU: 1500
  Dead timer due in 32 sec
  Retrans timer interval: 5
  Neighbor is up for 00h06m32s
  Neighbor Up Time : 2026-06-06 02:55:12
  Authentication Sequence: [ 0 ]

Area 0.0.0.0 interface 10.10.20.1 (Eth1/0/3)'s neighbors
Router ID: 3.3.3.3           Address: 10.10.20.2
  State: Full                Mode:Nbr is Master    Priority: 1
  DR: 10.10.20.1            BDR: 10.10.20.2        MTU: 1500
  Dead timer due in 39 sec
  Retrans timer interval: 5
```

Gambar 30: Ping dari VPCS VLAN 30 (IT) menuju gateway 192.168.30.1 dan ke 192.168.50.10 (PC LAN Huawei) — berhasil

```
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

 10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
C    10.10.10.0/30 is directly connected, GigabitEthernet0/3
L    10.10.10.1/32 is directly connected, GigabitEthernet0/3
C    10.10.11.0/30 is directly connected, GigabitEthernet0/0
L    10.10.11.1/32 is directly connected, GigabitEthernet0/0
O    10.10.20.0/30 [110/110] via 10.10.11.2, 00:07:25, GigabitEthernet0/0
      [110/110] via 10.10.10.2, 00:07:25, GigabitEthernet0/3
C    10.10.30.0/30 is directly connected, GigabitEthernet0/1
L    10.10.30.1/32 is directly connected, GigabitEthernet0/1
O    192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
```

Gambar 31: Ping dari VPCS VLAN 40 (Marketing) menuju 192.168.50.10 (PC LAN Huawei) dan ping antar-VLAN — berhasil

1.5 Praktikum 5: Pengujian *Failover* OSPF

Pada tahap ini, pengujian skenario kegagalan (*failover*) dilakukan menggunakan perintah `trace` untuk memvisualisasikan perpindahan jalur pengiriman paket ketika tautan utama mengalami gangguan.

1. Kondisi Normal (Dua Link Utama Aktif):

Saat kedua link Cisco–Huawei aktif, perintah `trace 192.168.50.10` dari Cisco Router menunjukkan trafik melewati jalur langsung Cisco → Huawei (*hop* pertama di jaringan 10.10.10.x atau 10.10.11.x).

```
Router>show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
  Known via "ospf 1", distance 110, metric 11, type intra area
  Last update from 10.10.11.2 on GigabitEthernet0/0, 00:03:45 ago
  Routing Descriptor Blocks:
    10.10.11.2, from 2.2.2.2, 00:03:45 ago, via GigabitEthernet0/0
      Route metric is 11, traffic share count is 1
    * 10.10.10.2, from 2.2.2.2, 00:03:45 ago, via GigabitEthernet0/3
      Route metric is 11, traffic share count is 1
Router>trace 192.168.50.10
Type escape sequence to abort.
Tracing the route to 192.168.50.10
VRF info: (vrf in name/id, vrf out name/id)
  1 10.10.10.2 8 msec
    10.10.11.2 4 msec
    10.10.10.2 3 msec
  2 192.168.50.10 9 msec 5 msec 5 msec
```

Gambar 32: Traceroute kondisi normal: trafik melewati jalur langsung Cisco → Huawei (10.10.10.2 / 10.10.11.2) dengan hanya 2 *hop*

2. Kondisi 1 Link Cisco–Huawei Mati:

Antarmuka `gi0/3` dimatikan menggunakan perintah `shutdown`. OSPF mendeteksi kegagalan dan mengkonvergensi. Hasil `trace` menunjukkan trafik tetap melewati jalur langsung Cisco–Huawei namun melalui antarmuka kedua (`gi0/0`, 10.10.11.x).

```

Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gi0/3
Router(config-if)# shutdown
Router(config-if)#exit
Router(config)#
*Jun 11 01:07:07.100: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on GigabitEthernet0/3 from FULL to DOWN, Neighbor Down: Interface down or detached
Router(config)#
*Jun 11 01:07:09.074: %LINK-5-CHANGED: Interface GigabitEthernet0/3, changed state to administratively down
*Jun 11 01:07:10.074: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/3, changed state to down
Router(config)#exit
Router#
*Jun 11 01:07:15.742: %SYS-5-CONFIG_I: Configured from console by console
Router#show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
  Known via "ospf 1", distance 110, metric 11, type intra area
  Last update from 10.10.11.2 on GigabitEthernet0/0, 00:07:13 ago
  Routing Descriptor Blocks:
    * 10.10.11.2, from 2.2.2.2, 00:07:13 ago, via GigabitEthernet0/0
      Route metric is 11, traffic share count is 1
Router#trace 192.168.50.10
Type escape sequence to abort.
Tracing the route to 192.168.50.10
VRF info: (vrf in name/id, vrf out name/id)
  1 10.10.11.2 4 msec 2 msec 7 msec
  2 192.168.50.10 8 msec 4 msec 5 msec

```

Gambar 33: Traceroute kondisi 1 link mati (*gi0/3 shutdown*): trafik berpindah ke link Cisco–Huawei kedua (10.10.11.2), konektivitas tetap terjaga

3. Kondisi 2 Link Cisco–Huawei Mati (*Failover* ke MikroTik):

Antarmuka kedua (*gi0/0*) ikut dimatikan sehingga tidak ada lagi jalur langsung Cisco–Huawei. OSPF segera konvergensi dan memindahkan seluruh lalu lintas ke jalur *backup* via MikroTik. Hasil *trace* menunjukkan jalur berubah menjadi Cisco → MikroTik (10.10.30.2) → Huawei → LAN.

```

Router#show ip route 192.168.50.0
*Jun 11 01:09:05.968: %SYS-5-CONFIG_I: Configured from console by console
Routing entry for 192.168.50.0/24
  Known via "ospf 1", distance 110, metric 201, type intra area
  Last update from 10.10.30.2 on GigabitEthernet0/1, 00:00:07 ago
  Routing Descriptor Blocks:
    * 10.10.30.2, from 2.2.2.2, 00:00:07 ago, via GigabitEthernet0/1
      Route metric is 201, traffic share count is 1
Router#trace 192.168.50.10
Type escape sequence to abort.
Tracing the route to 192.168.50.10
VRF info: (vrf in name/id, vrf out name/id)
  1 10.10.30.2 8 msec 2 msec 1 msec
  2 10.10.20.1 6 msec 3 msec 3 msec
  3 192.168.50.10 6 msec 9 msec 4 msec
Router#

```

Gambar 34: Traceroute kondisi *failover* (2 link Cisco–Huawei mati): jalur berubah melewati MikroTik (10.10.30.2 → 10.10.20.1) sebelum mencapai tujuan di Huawei

4. Mengaktifkan Kembali Link (*Fallback*):

Kedua antarmuka (*gi0/0* dan *gi0/3*) diaktifkan kembali menggunakan *no shutdown*. OSPF

kembali mendeteksi link yang pulih, melakukan konvergensi, dan memperbarui tabel routing. Trafik kembali dialihkan ke jalur utama Cisco–Huawei yang memiliki *cost* lebih rendah (*cost* 11) dibandingkan jalur via MikroTik (*cost* 201).

```
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gi0/3
Router(config-if)# no shutdown
Router(config-if)#exit
Router(config)#
Router(config)#interface gi0/0
Router(config-if)# no shutdown
Router(config-if)#exit
Router(config)#
*Jun 11 01:11:43.231: %LINK-3-UPDOWN: Interface GigabitEthernet0/3, changed state to up
*Jun 11 01:11:43.433: %LINK-3-UPDOWN: Interface GigabitEthernet0/0, changed state to up
*Jun 11 01:11:44.231: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/3, changed state to up
*Jun 11 01:11:44.433: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
Router(config)#show ip route 192.168.50.0
*Jun 11 01:11:50.460: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on GigabitEthernet0/0 from LOADING to FULL, Loading Done
*Jun 11 01:11:51.452: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on GigabitEthernet0/3 from LOADING to FULL, Loading Done
^
% Invalid input detected at '^' marker.

Router(config)#exit
Router#show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
  Known via "ospf 1", distance 110, metric 201, type intra area
  Last update from 10.10.30.2 on GigabitEthernet0/1, 00:02:58 ago
  Routing Descriptor Blocks:
    * 10.10.30.2, from 2.2.2.2, 00:02:58 ago, via GigabitEthernet0/1
      Route metric is 201, traffic share count is 1
Router#
*Jun 11 01:11:57.475: %SYS-5-CONFIG_I: Configured from console by console
Router#trace 192.168.50.10
Type escape sequence to abort.
Tracing the route to 192.168.50.10
VRF info: (vrf in name/id, vrf out name/id)
 0 10.10.10.2 6 msec
 1 10.10.11.2 3 msec
 2 10.10.10.2 3 msec
 3 192.168.50.10 11 msec 6 msec 4 msec
Router#
Router#show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
  Known via "ospf 1", distance 110, metric 11, type intra area
  Last update from 10.10.11.2 on GigabitEthernet0/0, 00:02:13 ago
  Routing Descriptor Blocks:
    10.10.11.2, from 2.2.2.2, 00:02:13 ago, via GigabitEthernet0/0
      Route metric is 11, traffic share count is 1
    * 10.10.10.2, from 2.2.2.2, 00:02:13 ago, via GigabitEthernet0/3
      Route metric is 11, traffic share count is 1
```

Gambar 35: Proses *fallback*: kedua link diaktifkan kembali, OSPF konvergensi, dan trafik kembali menggunakan jalur utama langsung Cisco → Huawei

2 Analisis Hasil Percobaan

Implementasi *VLAN Trunking* dengan arsitektur ROAS (*Router-on-a-Stick*) berjalan secara presisi. Protokol 802.1Q berhasil memberikan tanda identifikasi (*tagging*) pada setiap *frame* Ethernet sehingga Cisco Router mampu mendistribusikan paket ke *subnet* yang tepat melalui satu antarmuka fisik. Pemberian IP dinamis melalui layanan DHCP dari router juga berfungsi dengan mulus, terbukti dari keberhasilan VPCS VLAN 10 dan VLAN 20 mendapatkan IP secara otomatis, mengeliminasi kebutuhan penugasan IP manual pada klien berskala besar.

Dalam aspek *routing*, OSPF mendemonstrasikan interoperabilitas yang sangat baik di atas infrastruktur *multivendor*. Perbedaan sistem operasi (Cisco IOS, Huawei VRP, dan MikroTik RouterOS) tidak menghalangi pertukaran *Link State Advertisement* (LSA), sehingga *Link State Database* (LSDB) pada setiap perangkat berhasil tersinkronisasi sempurna hingga mencapai status *FULL* pada semua pasangan *neighbor*. Hal ini membuktikan bahwa OSPF sebagai protokol *open-standard* (RFC 2328) mampu beroperasi lintas vendor tanpa memerlukan konfigurasi tambahan yang bersifat vendor-spesifik.

Manipulasi parameter *cost* merupakan kunci dari rekayasa lalu lintas (*traffic engineering*) pada

percobaan ini. Dengan memberikan *cost* rendah (10) pada link langsung Cisco–Huawei dan *cost* tinggi (100) pada link via MikroTik, algoritma Dijkstra (*Shortest Path First*) terbukti memprioritaskan rute dengan kalkulasi metrik terendah sebagai jalur utama. Ketika kedua tautan fisik pada rute utama disimulasikan putus (*shutdown*), protokol OSPF dengan cepat mendeteksi ketiadaan paket *Hello* dari *neighbor* dan secara otonom memfasilitasi proses *failover* ke jalur MikroTik tanpa memutuskan konektivitas *end-to-end* dalam waktu yang signifikan. Begitu pula ketika link utama dipulihkan, mekanisme *fallback* berjalan otomatis dan trafik kembali menggunakan jalur optimal.

3 Hasil Tugas Modul

Pengerjaan Tugas Modul 5 mengenai “Implementasi Jaringan Enterprise HQ–Branch dengan VRRP, ISC-DHCP, FortiGate, GRE Tunnel, dan OSPF” telah dilaksanakan secara berkelompok. Solusi pengerjaan, skrip konfigurasi *firewall/tunneling*, dan dokumentasi pendukung telah diunggah ke repositori GitHub pada tautan berikut:

<https://github.com/hideaki7/Praktikum-Jarkom-2026-Kelompok-4/tree/main/MODUL%205>

4 Kesimpulan

Berdasarkan serangkaian percobaan yang telah dilakukan, dapat ditarik beberapa kesimpulan sebagai berikut:

- Integrasi VLAN dan *Trunking* melalui arsitektur ROAS secara efektif mengoptimalkan penggunaan antarmuka fisik router sekaligus menjaga keamanan dan isolasi *broadcast domain* antar-divisi. Konfigurasi subinterface dengan enkapsulasi 802.1Q memungkinkan satu antarmuka fisik melayani multiple VLAN sekaligus.
- Protokol OSPF adalah protokol *open-standard* yang sangat tangguh untuk mengelola *routing* dinamis pada topologi *multivendor*. Perbedaan platform (Cisco IOS, Huawei VRP, MikroTik RouterOS) tidak menghambat terbentuknya *adjacency* OSPF dan sinkronisasi *Link State Database*, menjamin konsistensi tabel *routing* di seluruh jaringan.
- Rekayasa nilai *cost* pada antarmuka OSPF merupakan mekanisme yang andal dalam membangun topologi *High Availability* (HA). Kemampuan konvergensi otomatis OSPF memastikan sistem memiliki toleransi kegagalan (*fault tolerance*) yang mumpuni: trafik secara otomatis berpindah ke jalur *backup* ketika jalur utama mengalami gangguan, dan kembali ke jalur optimal ketika jalur utama pulih, tanpa intervensi manual.

5 Lampiran

5.1 Dokumentasi saat Praktikum

Berikut adalah dokumentasi tambahan yang diambil selama pelaksanaan praktikum sebagai bukti pelaksanaan kegiatan.

